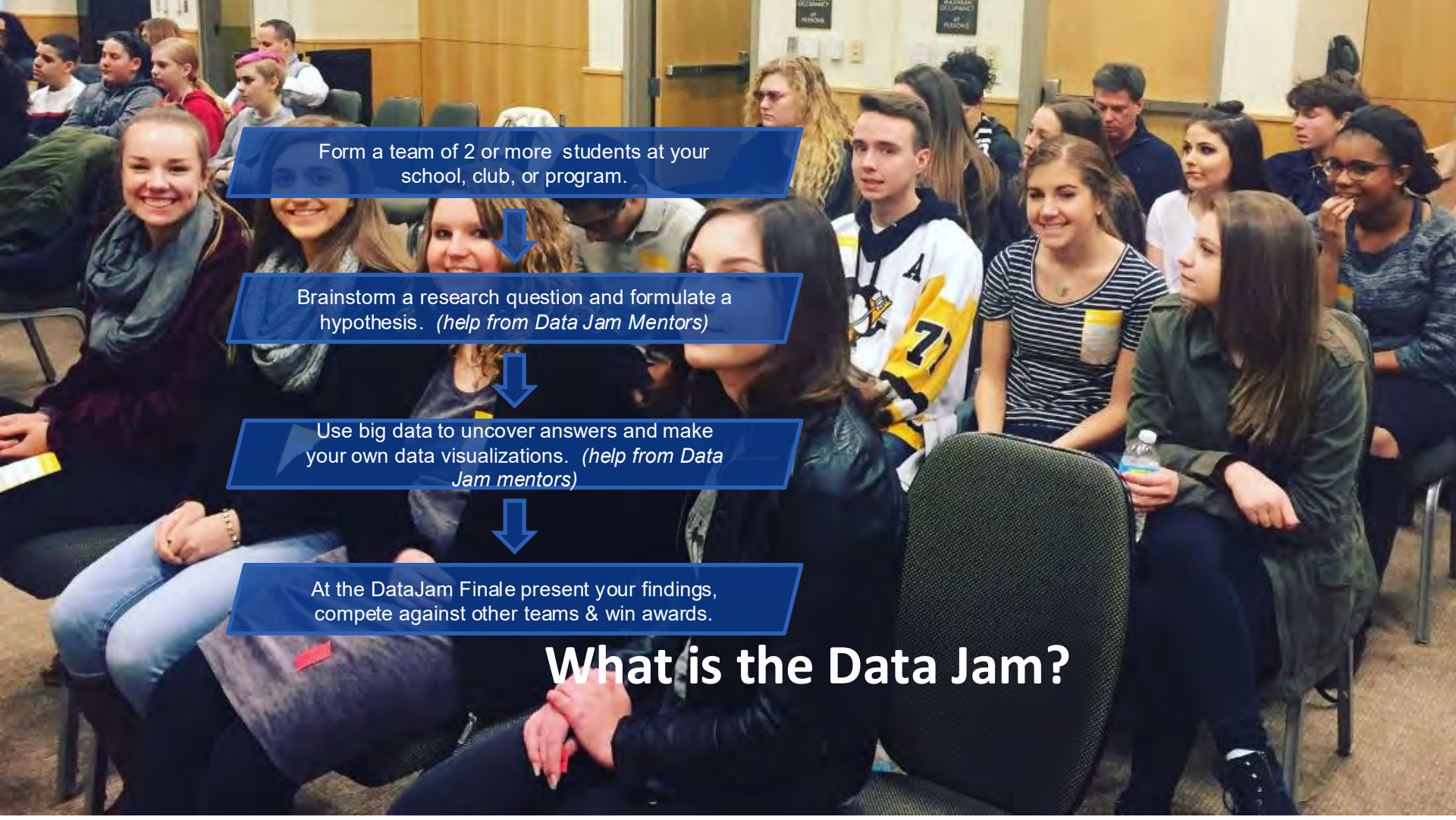


DataJam Advisor Workshop





Form a team of 2 or more students at your school, club, or program.

Brainstorm a research question and formulate a hypothesis. *(help from Data Jam Mentors)*

Use big data to uncover answers and make your own data visualizations. *(help from Data Jam mentors)*

At the DataJam Finale present your findings, compete against other teams & win awards.

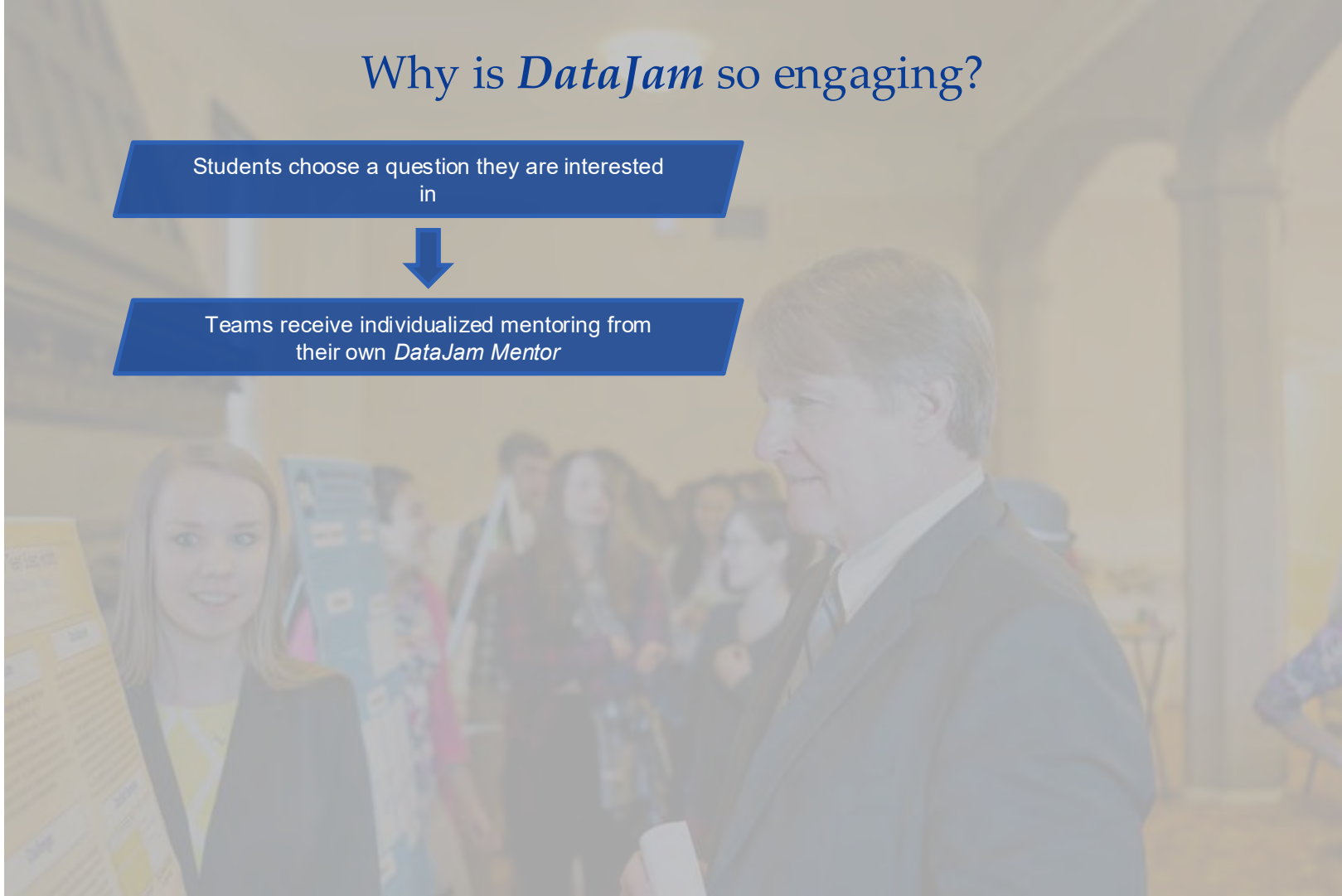
What is the Data Jam?

Why is *DataJam* so engaging?

Students choose a question they are interested in



Teams receive individualized mentoring from their own *DataJam Mentor*



The DataJam In

Brief



Excitement

A competition focusing on **the process and uses of big data** to **answer research questions**

Skills

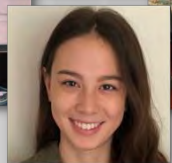
Formulate a research question **use case**., find **data**, **analyze** their data., make data **visualizations**., tell a **data story** and **present** findings to a panel of data science judges..



Resources

Free resources to High School students and afterschool programs., such as the Boys & Girls Clubs., **reduces barriers to Data Science career entry**..

The Mentors



Training

The DataJam Mentors are college students interested in the **power of data** and data **analysis** who are **trained** to work with **diverse** communities and needs..

Culture

DataJam Mentors volunteer their time., creating a lifetime journey of **community** and **inspiration** based on **trust** and new ways of working..



The DataJam Goals

By providing awareness, access and community to students, especially those with minimal resources, we all win!

Win-Win-Win



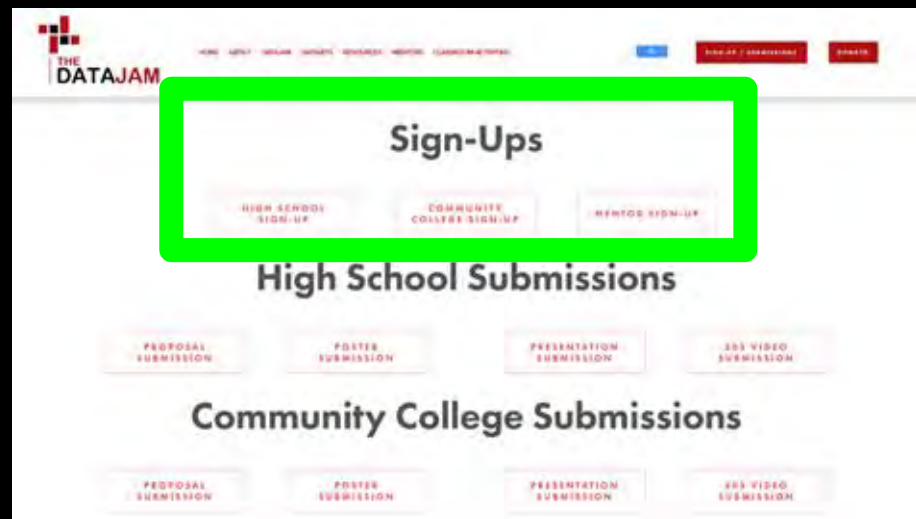
Find out more at thedatajam.org

How to Sign-Up your DataJam Team!

What you'll need:

- Student Names and eMails
- Team Number you assign to each team - even if there is only one
- Availability of Teams to meet with a Mentor

Let's do an example!





Adjusting Teams - Adding or Removing Students

Instead of filling out a new form,
simply email us your changes!
(datajam@thedatajam.org)



We understand the need for changes happens. If you think you will have many changes, please wait until you have several and let us know all at once.

Permission Slips - Filling them out and submitting them!

For high school students, permission slips must be filled out by each student and signed by a parent or guardian.

How permission slips are submitted is up to you! Permission slips can be submitted by each student or collected and combined and submitted by you. Completed permission slips are emailed to datajam@thedatajam.org.

Let's do one together!


THE DATAJAM

DataJam 2026

Parent Permission Slip, Waiver of Liability, Assumption of Risk, Indemnity Agreement

The DataJam (thedatajam.org) is the educational nonprofit that runs the High School DataJam each year to introduce high school students to learning about data science, how critical it is for understanding everything in our current world, and how to work with big data sets from a data visualization and statistics perspective.

We encourage all high schools to put together a DataJam team (2 or more students constitute a team, generally teams have 3-4 students). Each high school team is asked to have a teacher at their high school to serve as a DataJam Advisor. Teams are mentored by university students who are trained as DataJam Mentors in how to teach data science, best practices in mentoring, working with diverse communities, and how to give scientific presentations. Mentors are advised to use Zoom and Slack for mentoring and NOT share personal contact information. *From: youth on social media, or have direct contact with youth.

The DataJam will make available an online 5 modules for helping them think about the real data, how to make graphs and how to put together meeting with a mentor as they can get outside and students will be online and by videoconferencing single students.

The timeline for this year is (1) Fall 2025 the DataJam 2025, (2) December 5, 2025 - final teams work on their projects with help of their School DataJam Facilitator (2025 - name given).

Waiver: I agree to being permitted to participate in the DataJam, on any of the college including negligence and sexual misconduct 2026.

Assumption of Risk: Participation in DataJam involves the risk of injury or death. I understand that I am assuming the risk of injury or death.

Indemnification and Hold Harmless: I agree to hold DataJam harmless from any and all claims (including attorney's fees) arising out of my participation in the DataJam.

Severability: I further agree that this Waiver shall be in full and enforceable as permitted by law to have full legal force and effect.

I am the parent of _____ who is a current high school student at _____ High School. I give my permission for my child to participate in online videoconferences and use the Slack website to participate in the High School DataJam 2026. I also give permission for The DataJam to post pictures from videoconferences of my child participating in High School DataJam 2026 activities.

Acknowledgment of Understanding: I have read this Waiver of Liability, Assumption of Risk, and Indemnity Agreement, fully understood its terms, and understand that I am giving up substantial rights, including my right to sue. I understand that I am signing the agreement freely and voluntarily, and intend my signature to be a complete and unconditional release of all liability to the greatest extent allowed by law.

Parent Signature: _____ Date: _____

Printed Parent Name: _____

Student contact email address (a publicly available email address works best as many high schools block emails from outside organizations such as The DataJam): _____

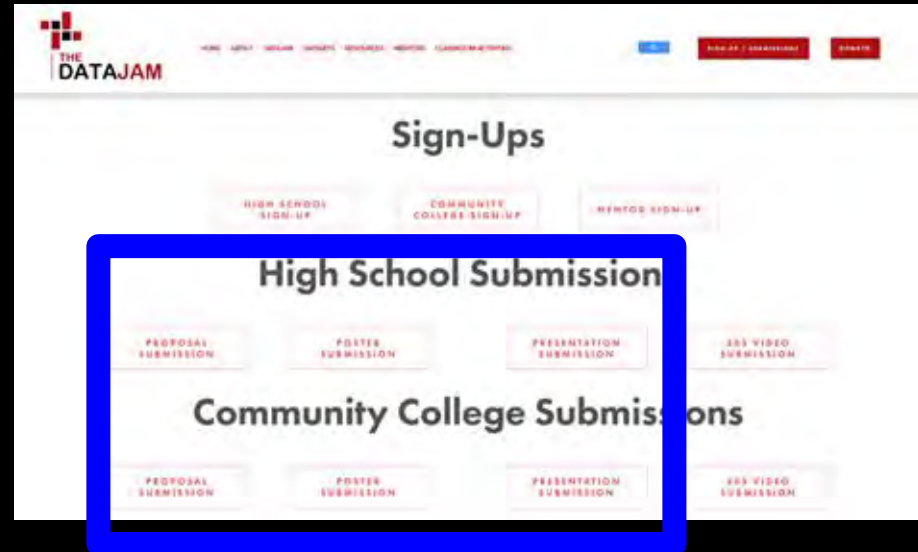
Seva and EMILIE TO @thedatajam@thedatajam.org

Proposals/Posters/Presentations

Submitting documents for the DataJam just got easier!

Now, teams or advisors submit their documents online! Each document should be submitted as a pdf.

We request that documents are filled out with your organization name and the teams name or number so we can quickly identify it.



Optimizing Work with Your Team's DataJam Mentor!

- DataJam Mentors are university students who have received specialized training to provide mentorship for the DataJam.
- Mentors will meet with your students by videoconference and using The DataJam SLACK workspace.
- It works best to arrange a regular time for the team to meet with the mentor.
- The mentor will provide regular summaries of their work with your students on a timeframe (e.g., weekly) agreed on by you and the mentor.
- Mentors can help with *every step* of The DataJam process.



Facilitating and Encouraging Teamwork!

- DataJam is a team competition. Teams *boost productivity, even out workload, allow for a deeper understanding of the topic, & make the DataJam experience more fun!*
- Teams should discuss ideas together so everyone is involved.
- Think about splitting up work.
- Progress checks and communication are essential!
- Set up regular meetings with your DataJam Mentor, and communicate between meetings on SLACK. Ask Mentors questions; if they don't know the answer they can seek guidance from The DataJam.

See the *Teamwork Guide* on the *Resources* page of The DataJam website.



Proposals and Initial Team Work

Coaching Strategies: We will discuss how to help students generate strong research questions, hypotheses, and analytical ideas.

1. Pick a topic close to "home"



Keep it
SIMPLE

2. Is this idea too broad?

3. Does this idea have data available?



Completed High School DataJam Projects

2016

2017

2018

2019

2020

2021

2022

2023

2024

2025

2025 POSTERS

Avonworth Team 1

The American Dream: Civic Engagement Across the White-Picket Fence

Avonworth Team 2

Dollars and Sense: The SAT-Income Equation

Avonworth Team 3

Moving Around the 'Burgh

Bisbee Team 1

Beaver Data Poster

Bisbee Team 2

Water Scarcity, Environmental Factors, and the Beaver Populations in the Sand Pedro River

Bisbee Team 3

The Migration of the Sand Pedro Beavers and its Effects on the Mexican Side

Canyon Crest Academy Team 6

Erasing the Unwanted: The Link Between Memory Suppression and Psychological Disorders

Canyon Crest Academy Team 7

EVALUATING THE DISPARITY BETWEEN COVID-19 VACCINE ADMINISTRATION RATES IN U.S. COUNTIES WITH LOW VS. HIGH MEDIAN HOUSEHOLD INCOMES

Canyon Crest Academy Team 8

Examining Racial Disparities in Health Care Access Specifically Whether the Individual is Insured or Uninsured Within Cities

Canyon Crest Academy Team 9

Correlation between AI Adoption Level and Job Industry

Canyon Crest Academy Team 10

Keystone Oaks Team 1

The Wizard of Oz(Empic)

Keystone Oaks Team 2

Class Size Beats Cash Prize

Maple Grove Team 1

How Coffee & Energy Drinks Affect YOUR Health

Monarch

Does the Population of Different Neighborhood Effect the Collision Rates in San Diego?

Moon Area High School

AI vs. Reality: Detecting the Markers of AI-Generated Images

Miami Valley Career Technology Center Team 1

Saturn's Moons

Passaic Academy Team 2

How does perceived ease of getting drugs vary from race?

Passaic Academy Team 3

Drug overdoses and unemployment in the New York

Passaic Academy Team 4

Can you be victim to nicotine?

Passaic Academy Team 5

Price of Fast Food vs Median Household Income per County in the US

Pittsburgh Science and Technology Academy

Does Pollution Affect the Organisms Around it?

Preuss Team 1

Is the cost of acquiring an undergraduate degree in the United States growing at a

Proposals and Initial Team Work

Agreeing on a Research

Question: This is one of the most difficult steps in a DataJam Project. It requires teamwork, time and investigation.

1. Suggest that each student on a team writes down 3 ideas for a research question.
2. Discuss each idea:
 - a. Is it a question that can be answered by data analysis?
 - b. Is it of interest to the whole team?
 - c. Will the answer be meaningful?
 - d. Will a dataset be publicly available to answer the question?

Proposals and Initial Team Work

Finding Data: Guidance on finding and getting data from the web using resources like the DataJam website and data.gov.

1. Check the DataJam website's Datasets page
2. Do a quick google search
3. Ask AI like ChatGPT - phrasing is important

Let's take a tour through the Datasets page.

Resources on The DataJam Website for getting started with a DataJam Proposal!

How to Conduct a DataJam Project

This brief overview will provide a summary of each of the six steps in the process of doing a DataJam Project.



data. Begin

searching for data to see if your top idea is a plausible one to pursue. If not, move down your rankings to see if your other ideas are valid. You find one that is.

- 2. Writing your Research Question & Hypothesis** - This is often the section students struggle the most with, as it can be difficult to word your questions in a way that it will allow you to fully explore the topic you are interested in. Writing up your research question and consulting your hypothesis is truly a process that takes time. We recommend first brainstorming questions, then confining with a narrow for revision. Then refine and repeat until these are worded in a way that they truly allow you to do the analyses you would like to do and you get the OK from the advisory board when you turn your proposal in.



- 3. Collecting and Preparing Data** - This section is where you finally start diving into the available data and compiling it! We recommend using either Excel or Google Sheets for

WRITING A DATA JAM RESEARCH QUESTION

Nikhil Srivastava, 2021 Data Jam Member

1 Choose Your Topic Of Interest

Brainstorm a list of topics that you find interesting and that you think could be represented with numerical data.

You don't need to have a specific question in mind at this point, just general themes and topics that you would like to explore.

The more you care about your topic, the more rewarding the research process will be.

2 Narrow Your Scope

Pick a few topics from your list that stand out to you. For each of them, begin to list specific subcategories that are more easily measurable.

If you are not extremely knowledgeable in the subject, this is a good time to do some background research on your topic in order to help you break it down into more distinct variables.

HOW TO DE-STRESS THE DATAJAM

The following advice forms have been taken from Tony Roddy's "Student Manual" in the section titled "How to Conduct a Data Jam Project at Business". Click [here](#) for the manual.

Click [here](#) to watch an introductory video to the DataJam

1. EXPLORING TOPICS

- Give everyone a vote at the start.
- There are 100 total votes.
- Having multiple topics to vote on ensures given your team the best possibility.

2. NARROWING YOUR SCOPE

- Have each team member come up with a potential question.
- Make the question specific.
- Click here for the 100 total votes.
- Form a research question with examples.

3. COLLECTING & PREPARING DATA

- Try to find datasets that are categorized properly.
- Click here for links to resources to search for datasets.

4. ANALYZING YOUR DATA

- Use Excel or Google Sheets to do your analysis.
- Click here for a Google Sheets tutorial.
- Click here for a Google Sheets tutorial.
- Click here for a Google Sheets tutorial.

5. PRESENTING YOUR DATA

- Click here to watch a video on presenting data with visuals.
- Click here for a Google Sheets tutorial.
- Click here for a Google Sheets tutorial.

6. INTERPRETING DATA AND FORMING CONCLUSIONS

- Be confident! You want to feel a story about your data.
- Create a story about your data and your project.
- Explain why your research question is important.
- Click here for an example of a "Conclusion and Recommendation".

1. Identify a
question

1. Find data

1. Practice filling
out the proposal
sheet.

Proposals and Initial Team Work

Your Task: You will create a brief, outline-level mini-DataJam proposal to practice these skills.

Experience a “Mock DataJam” as a Team

We will now break out into group(s) and practice interacting with a mentor during each stage of The DataJam competition.



Regroup and Final Thoughts

- What was the most challenging part of creating your mini-proposal?
- Based on your experience today, what do you anticipate will be the biggest challenge for your student teams?
- What's one new idea or piece of information you'll take away from this workshop that you're excited to apply with your team?
- Why is it so important to have students define their research question before they start looking for data?